

COSC 594/650 — Homework 1 Walkthrough

(Study notes, not a submission)

Problem 1 — Convex optimization

Minimize $e^{x_1} + e^{x_2} + \cdots + e^{x_n}$ subject to $x_1 + x_2 + \cdots + x_n = \gamma$.

(a) Convexity

Each term e^{x_i} is the composition of the convex, increasing function $t \mapsto e^t$ with the affine map $x \mapsto x_i$, hence convex; a sum of convex functions is convex. The constraint set $\{x : \sum_i x_i = \gamma\}$ is an affine subspace, hence convex. Minimizing a convex function over a convex set is a convex problem. ■

(b) Eliminating the constraint

Substituting $x_n = \gamma - x_1 - \cdots - x_{n-1}$ gives the unconstrained problem in $(x_1, \dots, x_{n-1}) \in \mathbb{R}^{n-1}$:

$$h(x_1, \dots, x_{n-1}) = e^{x_1} + \cdots + e^{x_{n-1}} + e^{\gamma - x_1 - \cdots - x_{n-1}}.$$

(c) Global optimum

h is still a sum of exponentials of affine functions, hence convex, so any stationary point is global. Setting $\partial h / \partial x_i = 0$ for $i = 1, \dots, n-1$:

$$e^{x_i} - e^{\gamma - x_1 - \cdots - x_{n-1}} = e^{x_i} - e^{x_n} = 0 \implies x_i = x_n.$$

So all coordinates are equal at the stationary point; combined with $\sum_i x_i = \gamma$, this gives $x_i = \gamma/n$ for all i . This is the unique global minimizer, with optimal value $ne^{\gamma/n}$.

(d) AM–GM inequality

Given any $(y_1, \dots, y_n) \in \mathbb{R}_+^n$, set $x_i = \ln y_i$ and $\gamma := \sum_i \ln y_i = \ln(\prod_i y_i)$. This x is feasible for the problem in (a)–(c). Since the global minimum value of $\sum_i e^{x_i}$ over that feasible set is $ne^{\gamma/n}$, feasibility of this x gives

$$\sum_i y_i = \sum_i e^{x_i} \geq ne^{\gamma/n} = n \left(\prod_i y_i \right)^{1/n}.$$

Dividing by n :

$$\frac{1}{n} \sum_{i=1}^n y_i \geq \left(\prod_{i=1}^n y_i \right)^{1/n}, \quad \forall (y_1, \dots, y_n) \in \mathbb{R}_+^n.$$

This is AM–GM, obtained directly from the result of (a)–(c). ■

Problem 2 — Convexity proofs

(a)

Let $f(x) = \log \left(e^{a_1^T x + b_1} + \dots + e^{a_n^T x + b_n} \right) = g(Ax + b)$, where $g(z) = \log \sum_i e^{z_i}$ is the *log-sum-exp* function and A has rows a_i^T . Composing a convex function with an affine map preserves convexity, so it suffices to show g is convex.

Let $p(z)$ be the softmax of z : $p_i(z) = e^{z_i} / \sum_j e^{z_j}$. Then

$$\nabla g(z) = p, \quad \nabla^2 g(z) = \text{diag}(p) - pp^T.$$

For any $v \in \mathbb{R}^n$,

$$v^T \nabla^2 g(z) v = \sum_i p_i v_i^2 - \left(\sum_i p_i v_i \right)^2 = \text{Var}_p(v) \geq 0,$$

treating p as a probability distribution over the values $v_1, \dots, v_n$ (this is exactly $\mathbb{E}_p[v^2] - (\mathbb{E}_p[v])^2 \geq 0$). So $\nabla^2 g \succeq 0$ everywhere, g is convex, and hence f is convex. ■

(b)

Let $\sigma_1(Z)$ denote the largest singular value of Z (the spectral norm), a matrix norm and hence convex and nonnegative. The function $g(X) := \sigma_1(AX)$ is convex in X , being the composition of the convex function $\sigma_1(\cdot)$ with the linear map $X \mapsto AX$; and $g(X) \geq 0$ always.

Write $f(X) = h(g(X))$ with $h(t) = \exp(t^{10})$. We only need h restricted to $[0, \infty)$, the range of g . There,

$$h'(t) = 10t^9 \exp(t^{10}) \geq 0 \quad \text{for } t \geq 0,$$

so h is convex and nondecreasing on $[0, \infty)$. By the composition rule (h convex nondecreasing on the range of g , g convex $\implies h \circ g$ convex), f is convex. ■

Problem 3 — Stationary points

Let $f(x, y) = x^2 + y^2 + \beta xy + x + 2y$ on $\mathbb{R}^2$. Setting $\nabla f = 0$:

$$2x + \beta y = -1, \quad \beta x + 2y = -2.$$

This is a linear system with matrix $M = \begin{pmatrix} 2 & \beta \\ \beta & 2 \end{pmatrix}$, $\det M = 4 - \beta^2$. Since f is quadratic, its Hessian is exactly M (constant), with eigenvalues $2 + \beta$ and $2 - \beta$ (eigenvectors $(1, 1)$ and $(1, -1)$, since the diagonal entries are equal).

Case $|\beta| < 2$. $\det M \neq 0$; by Cramer's rule,

$$x^* = \frac{2(\beta - 1)}{4 - \beta^2}, \quad y^* = \frac{\beta - 4}{4 - \beta^2}.$$

Both eigenvalues $2 \pm \beta$ are positive, so $M \succ 0$ everywhere, f is strictly convex, and (x^*, y^*) is the unique **global minimum**.

Case $\beta = 2$. The system becomes $2x + 2y = -1$ and $2x + 2y = -2$ — inconsistent, so **no stationary points exist**. Consistently, f is unbounded below: along $(x, y) = (t, -t)$,

$$f(t, -t) = 2t^2 - 2t^2 + t - 2t = -t \longrightarrow -\infty \text{ as } t \rightarrow \infty.$$

Case $\beta = -2$. Similarly inconsistent; along (t, t) , $f(t, t) = 3t \rightarrow -\infty$ as $t \rightarrow -\infty$. No stationary points.

Case $|\beta| > 2$. $\det M \neq 0$ again, so the same point (x^*, y^*) from Cramer's rule is stationary — but now one eigenvalue of M is negative, so the Hessian is indefinite: this is a **saddle point**, not a local minimum. f is unbounded below along the negative-eigenvalue direction, so no global minimum exists.

Summary. A global minimum exists — and is the unique stationary point — exactly when $|\beta| < 2$. For $|\beta| \geq 2$, $\inf f = -\infty$.

Problem 4 — Linewise local minima

(a)

Fix $d \in \mathbb{R}^n$ and let $g(\alpha) = f(x^* + \alpha d)$. By the chain rule, $g'(\alpha) = \nabla f(x^* + \alpha d)^T d$, so $g'(0) = \nabla f(x^*)^T d$. Since g has a local minimum at the interior point $\alpha = 0$ and f is differentiable, $g'(0) = 0$. This holds for every $d \in \mathbb{R}^n$, so $\nabla f(x^*)^T d = 0$ for all d ; taking $d = \nabla f(x^*)$ gives $\|\nabla f(x^*)\|^2 = 0$, i.e. $\nabla f(x^*) = 0$. ■

(b)

Let $f(y, z) = (z - py^2)(z - qy^2)$ with $0 < p < q$. Consider $(0, 0)$.

Along any line $(y, z) = (\alpha a, \alpha b)$:

$$g(\alpha) = f(\alpha a, \alpha b) = (\alpha b - p\alpha^2 a^2)(\alpha b - q\alpha^2 a^2).$$

- If $a = 0$: $g(\alpha) = (\alpha b)^2 \geq 0$, a strict global minimum at $\alpha = 0$.
- If $b \neq 0$: $g(\alpha) = \alpha^2[(b - p\alpha a^2)(b - q\alpha a^2)]$, and the bracket $\rightarrow b^2 > 0$ as $\alpha \rightarrow 0$, so it stays positive for small α . Hence $g(\alpha) > 0 = g(0)$ near 0: a strict local minimum.
- If $b = 0$, $a \neq 0$: $g(\alpha) = pq a^4 \alpha^4 \geq 0$, minimized at $\alpha = 0$.

So $(0, 0)$ is a strict local minimum of f along *every* line through it.

But $(0, 0)$ is *not* a local minimum of f overall. Take $p < m < q$ and the parabola $z = my^2$:

$$f(y, my^2) = (my^2 - py^2)(my^2 - qy^2) = y^4(m - p)(m - q).$$

Since $m - p > 0$ and $m - q < 0$, this is strictly negative for every $y \neq 0$, while $f(0, 0) = 0$. As $y \rightarrow 0$, the points $(y, my^2) \rightarrow (0, 0)$ while $f < 0$ there — so every neighborhood of the origin contains points with f strictly less than $f(0, 0)$. Hence $(0, 0)$ is **not** a local minimum, despite appearing to be one along every straight line: the parabola is tangent to every line through the origin, so no single line detects the dip.

Problem 5 — Symmetric rank-one matrix approximation

Let $A = x_0 x_0^T \neq 0$. Expanding,

$$f(x) = \frac{1}{2} \|A - xx^T\|_F^2 = \frac{1}{2} \|A\|_F^2 - x^T A x + \frac{1}{2} \|x\|^4,$$

using $\text{tr}(Axx^T) = x^T A x$ and $\|xx^T\|_F^2 = \text{tr}(xx^T xx^T) = \|x\|^4$.

(a) Convexity

No. Along $x = su_0$ with $u_0 = x_0/\|x_0\|$:

$$f(su_0) = \frac{1}{2} \|A\|_F^2 - s^2 \|x_0\|^2 + \frac{1}{2} s^4,$$

whose second derivative at $s = 0$ is $-2\|x_0\|^2 < 0$: f is locally concave there, so it is not convex.

(b) Stationary points

$\nabla f(x) = -2Ax + 2\|x\|^2 x = 2(\|x\|^2 I - A)x$. Setting this to zero:

$$Ax = \|x\|^2 x,$$

so $x = 0$, or x is an eigenvector of A with eigenvalue $\|x\|^2$. Since $A = x_0 x_0^T$ has eigenvalue $\|x_0\|^2$ (eigenvector x_0) and eigenvalue 0 with multiplicity $n - 1$ (eigenspace $x_0^\perp$):

- In the 0-eigenspace: $Ax = 0$ forces $\|x\|^2 x = 0 \Rightarrow x = 0$.
- Along x_0 : $x = cu_0$ needs $c^2 = \|x_0\|^2 \Rightarrow c = \pm\|x_0\|$, giving $x = \pm x_0$.
- Vectors with both an x_0 -component and an orthogonal component cannot satisfy the equation unless the orthogonal part vanishes.

Stationary points: $x = 0$, $x = x_0$, $x = -x_0$.

(c) Local vs. global optima

$f(x_0) = f(-x_0) = \frac{1}{2} \|x_0\|^4 - \|x_0\|^4 + \frac{1}{2} \|x_0\|^4 = 0$. Since $f \geq 0$ always with equality iff $xx^T = A$, these achieve the **global minimum** (value 0).

$f(0) = \frac{1}{2} \|A\|_F^2 = \frac{1}{2} \|x_0\|^4 > 0$. Along $x = \varepsilon u_0$: $f = \frac{1}{2} \|x_0\|^4 - \varepsilon^2 \|x_0\|^2 + O(\varepsilon^4)$, which *decreases* as ε moves away from 0 — so $x = 0$ is a **saddle point**, not a local minimum.

The only local minima are $\pm x_0$, and both are global. **Yes: every local optimum is a global optimum.** (This is the classical “no spurious local minima” landscape for rank-one PCA-type factorization.)

Problem 6 — General rank-one factorization

Let $A = x_0 y_0^T \neq 0$ (not necessarily symmetric). Expanding,

$$f(x, y) = \frac{1}{2} \|A\|_F^2 - x^T A y + \frac{1}{2} \|x\|^2 \|y\|^2.$$

(a) Convexity

No. Even in the scalar case $n = 1$, $f(x, y) = \frac{1}{2}(a - xy)^2$ is a “double-well” quartic along $x = y = t$, manifestly nonconvex.

(b) Stationary points

$$\nabla_x f = -Ay + \|y\|^2 x = 0, \quad \nabla_y f = -A^T x + \|x\|^2 y = 0.$$

With $A = x_0 y_0^T$: $Ay = (y_0^T y)x_0$ and $A^T x = (x_0^T x)y_0$, so the conditions become

$$(y_0^T y)x_0 = \|y\|^2 x, \quad (x_0^T x)y_0 = \|x\|^2 y.$$

- If $x = 0$: the second equation is automatic; the first forces $y_0^T y = 0$, i.e. $y \perp y_0$. So $\{x = 0, y \perp y_0\}$ are all stationary.
- Symmetrically, $\{y = 0, x \perp x_0\}$ are all stationary.
- If $x, y \neq 0$: the equations force $x \parallel x_0$ and $y \parallel y_0$ (otherwise one side has a component the other lacks). Writing $x = \lambda x_0$, $y = \mu y_0$ and substituting back, both equations reduce to $\lambda\mu = 1$. So for any $\lambda \neq 0$, $(x, y) = (\lambda x_0, y_0/\lambda)$ is stationary — and

$$xy^T = \lambda x_0 \cdot \frac{1}{\lambda} y_0^T = x_0 y_0^T = A \quad \text{exactly,}$$

so $f = 0$ on this entire family: every such point is already a perfect factorization.

Stationary set: $\{x = 0, y \perp y_0\} \cup \{y = 0, x \perp x_0\} \cup \{(\lambda x_0, \lambda^{-1} y_0) : \lambda \neq 0\}$.

(c) Every local optimum is global

The third family already achieves $f = 0$, the global minimum (since $f \geq 0$ with equality iff $xy^T = A$) — trivially global optima.

Consider a degenerate stationary point $(0, \bar{y})$ with $y_0^T \bar{y} = 0$, where $f(0, \bar{y}) = \frac{1}{2}\|A\|_F^2 > 0$. Perturb $x = \varepsilon x_0$, $y = \bar{y} + \delta y_0$:

$$f(\varepsilon x_0, \bar{y} + \delta y_0) = \frac{1}{2}\|A\|_F^2 - \varepsilon\delta\|x_0\|^2\|y_0\|^2 + \frac{1}{2}\varepsilon^2\|x_0\|^2(\|\bar{y}\|^2 + \delta^2\|y_0\|^2),$$

using $y_0^T \bar{y} = 0$ to kill a cross term. Set $\delta = k\varepsilon$ for a constant k large enough that

$$-k\|x_0\|^2\|y_0\|^2 + \frac{1}{2}\|x_0\|^2\|\bar{y}\|^2 < 0$$

(always achievable). Then for small $\varepsilon > 0$, $f(\varepsilon x_0, \bar{y} + k\varepsilon y_0) = \frac{1}{2}\|A\|_F^2 + (\text{negative})\varepsilon^2 + O(\varepsilon^3)$, strictly less than $f(0, \bar{y})$. So $(0, \bar{y})$ is not a local minimum — it is a saddle. The symmetric argument handles $\{y = 0, x \perp x_0\}$.

Every stationary point is therefore either in the global-minimizer family (value 0) or a saddle (not a local minimum at all). Hence **every local optimum is a global optimum**. ■ This is the archetypal “benign nonconvex landscape” result underlying why gradient-based methods succeed on matrix factorization problems despite nonconvexity.